# UML Class Diagram Report 2

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Project Title: Coffee Shop Management System

## 1. System Overview

This report presents the structural design for the Coffee Shop Management System using a UML Class Diagram. The proposed system models four main entities: Customer, MenuItem, Order, and Payment.

## 2. Classes, Attributes, and Methods

### Class 1: Customer

- Description: Represents the customer who places orders within the system.
- Attributes:
  - customerID : String
  - name : String
  - email : String
  - phone : String
- Methods:

  - placeOrder() : void

  - updateProfile() : void

Class 2: MenuItem

- Description: Represents items available for purchase on the coffee shop menu.

- Attributes:

  - itemID : String

  - name : String

  - description : String

  - price : double

  - category : String

- Methods:

  - getDetails() : String

  - updatePrice() : void

Class 3: Order

- Description: Handles customer orders, tracks order statuses, and calculates costs.

- Attributes:

  - orderID : String

  - orderDate : Date

  - status : String

  - totalPrice : double

- Methods:

  - addItem() : void

  - calculateTotal() : double

  - updateStatus() : void

Class 4: Payment

- Description: Manages payment transactions and processing for completed orders.

- Attributes:

  - paymentID : String

  - amount : double

  - paymentDate : Date

  - paymentMethod : String

- Methods:

  - processPayment() : void

  - refund() : void

3. Class Relationships and Multiplicities

- Customer to Order (Association): 1 Customer to * Orders. A single customer can place one or multiple orders over time, while each order belongs to one specific customer.

- Order to MenuItem (Aggregation / Association): 1 Order contains * MenuItems. An order consists of one or more menu items. A menu item can appear across multiple customer orders.

- Order to Payment (Association): 1 Order to 1 Payment. Each completed order is associated with exactly one payment transaction.